

Canoe VR: An Immersive Exergame to Support Cognitive and Physical Exercises of Older Adults

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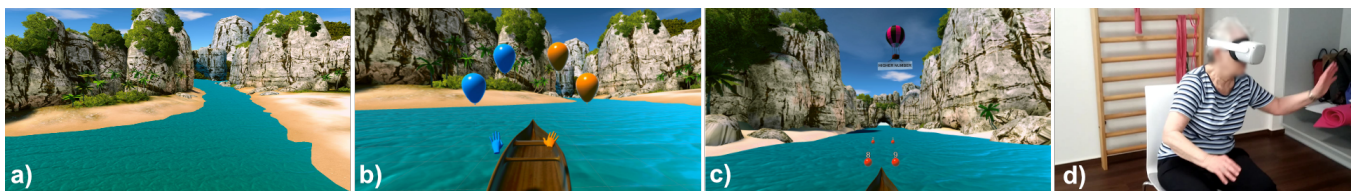


Figure 1: The figure shows the last version of *Canoe VR*: a) The game environment. b) Players calibrate their arm reach by pushing balloons away. c) Players pop soap bubbles with a higher number. d) An older adult plays the game.

ABSTRACT

Cognitive and physical exercises are important factors to support a healthy life, especially considering demographics change. Virtual reality (VR) exergames have great potential to support these activities in a more motivating way. However, regular usage of VR exergames by the older population is still limited. To address this issue, we designed and implemented a VR exergame: *Canoe VR*. We applied several prototyping sessions with older players and also report the results from interviews with physiotherapists. The results suggest that *Canoe VR* was very well received and can be used by older players as an additional fitness tool. We discuss the implications of extending a fitness routine with a VR exergame and using the game with players of different abilities.

CCS CONCEPTS

• **Software and its engineering** → **Interactive games**; • **Human-centered computing** → **Accessibility**.

KEYWORDS

Virtual reality, exergames, older adults

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1 INTRODUCTION

Given that the world population is aging [34], performing cognitive [35] and physical [8] exercises holds major importance for maintaining a healthy life. These exercises can help to reduce age-related decline in abilities [8, 21] and support well-being [7, 27]. Despite these benefits, having no interest, lack of opportunities for physical exercises, or COVID-19 restrictions can negatively affect participating in these activities [24, 29].

Immersive virtual reality (VR) systems are becoming commercially available and increasingly affordable for consumers (e.g., Meta Quest 2 [23]). In recent years, several VR exergames [37] (such as Beat Saber [13], OhShape [19], BoxVR [11]) have been introduced that incorporate physical exercises with cognitive aspects. Such training games can motivate users to exercise and continue training over longer periods. Especially given the effects of COVID-19, these systems allow exercising at home without contact with others or to reach exercise services without traveling. Nevertheless, VR technology is still novel and can expose some difficulties for users such as usability problems [1], cybersickness [20] or ergonomics [15]. At the same time, designing these games for older generations requires attention to the game design, age-related changes, and movements employed in the game.

In this paper, we implemented *Canoe VR* following a human-centered design (HCD) approach [12]. Older adults, fitness experts,

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Figure 2: The figure shows *a)* an artificial agent playing the game together with the user. *b)* Two older adults in the multiplayer version. *c)* The representation of the two players in the multiplayer version. *d)* A physiotherapist testing the game.

and a health professional were involved in the process. We conducted eight prototyping sessions that considered nuances in the older population, such as older adults with and without dementia. The game was also tested in a gym with older adults and fitness experts. Based on the gathered feedback of play-testings and interviews, *Canoe VR* has been iteratively improved. We report the results of these sessions and lessons learned, which can guide future work on VR exergames.

The contribution of this paper are:

- Demonstration of a VR exergame designed for older users by considering nuances in the population as well as their training needs,
- Use of this VR exergame in the regular training sessions of older adults as an additional fitness tool.

2 RELATED WORK

Previous work showed that cognitive [35] and physical [8] activities are beneficial for healthy aging. For example, they can support cognitive functions [35] and reduce the effects of aging in the brain [8]. It is suggested that the combination of both interventions can lead to better outcomes [10], and in particular simultaneous training, can benefit cognitive functions of older population [32].

The World Health Organization recommends that older adults should be physically active (at least 150-300 minutes of moderate activity per week) [39]. However, COVID-19 imposed some challenges to obtain these activities due to necessary lockdown policies. While prior work identified some obstacles to physical activities (e.g., poor health conditions and missing company) [24] for older adults, the impacts of COVID-19 on physical activity is inevitable [29, 40]. Relating to our paper, we argue that exergames can offer simultaneous cognitive and physical training possibilities, in particular, during the COVID-19 pandemic. Thus, they can be a supportive tool for exercise routines of older adults.

The use of immersive VR is increasing [2] and it is becoming common for several applications ranging from business to gaming industry. VR supports unique experiences such as a potentially intense sense of presence [30], multisensory sensations [31], and natural interactions. For instance, in VR, older adults can experience outdoor world which might be limited due to COVID-19 and their health condition [17].

With VR exergames, players can benefit from both physical and cognitive stimulation while enjoying the game experience. In recent years, VR exergames were also explored for older adults with [16] and without dementia [18]. In an HCD process, Karaosmanoglu et al. [16] designed a VR exergame (*Memory Journalist VR*) for older

players with dementia that included multiple stakeholders in the process, indicating positive player experiences for users. Similarly, a recent work [25] presented a VR exergame (*Seas the Day*) designed with input from various stakeholders for people with dementia. Kruse et al. [18] compared a VR exergame (*Maestro Game VR*) to a video-guided exercises in older adults. The findings suggested that both training options were comparably similar in terms of enjoyment and attention, and VR exergames can be a good addition for training. A recent study [6] indicated positive outcomes (e.g., usability, positive experiences) for the use of commercial VR games (e.g., BoxVR [11]) with four older adults during the pandemic.

In our work, we consider varying user groups (i.e., with and without dementia) in a single game and design for these groups, and use this VR exergame as an additional fitness tool. In this regard, users' needs, age-related changes, and health conditions should be considered [12, 14, 16]. Thus, in this paper, we used the HCD process [12] that considers needs and preferences of users by involving end-users as well as other stakeholders.

3 CANOE VR

In this section, we describe the technical properties and the last version of the game design. The changes applied to the game over the prototyping sessions can be found in Table 2.

The game was implemented in Unity (v. 2019.4.18f1) with the Universal Render Pipeline [33]. Since we used the Valve Index [36] in the early sessions, the game was run with an i9-9900K CPU and a GeForce RTX2080 Ti graphics card. Later, we used the Meta Quest 2 [23] as it is a standalone device and allows to conducting field studies without being bound to a computer. With this device, we were able to easily integrate hand tracking into our game.

The game play takes place on a tropical island scene (see Figure 1a), adapted from *BoatAttack* [22, 28]. Players are positioned inside a canoe that moves down a river. The game starts with a calibration phase. First, players perform a thumbs up gesture with both hands or they can press a button on both of their controllers to calibrate their seating position and comfortable view direction. Then, to obtain the maximum arm range of motion (ROM), the players need to touch four balloons (left, right, left up, right up) and push them away as far as possible (see Figure 1b).

After this calibration, users can select their game level. The game currently offers 22 levels with varying cognitive and physical tasks (see Table 1). Moreover, players can adjust the difficulty, length, and the number of game levels.

When the player presses a start button, the game provides written and audio instructions describing the level. Also as a reminder,

Table 1: This table lists the game levels in *Canoe VR*. The asterisk indicates levels that can only be played in multiplayer mode.

GAME LEVELS	DESCRIPTION	GAME LEVELS	DESCRIPTION
(1) All Good	Touch as many bubbles as possible.	(12) Hand Change	Collect the bubbles by changing hands.
(2) Only Colored	Touch all the bubbles except the black ones.	(13) Fruit	Collect all fruits.
(3) Only Red	Collect only red bubbles.	(14) Music	Collect the bubbles with accompanying music.
(4) Two Simultaneously	Collect two bubbles at the same time.	(15) Dividable by 7	Collect all the bubbles showing a number dividable by seven.
(5) Color Change	Collect a red bubble and then a yellow bubble alternately.	(16) Lower Number	Collect the bubble with the lower number.
(6) Higher Number	Collect the bubble with the higher number.	(17) Vowels	Collect the bubbles showing a vowel.
(7) Maximum 5 Yellow	Collect a maximum of 5 yellow bubbles, but all red bubbles.	(18) Letters in Word	Collect all the bubbles showing letters contained in the word "canoe".
(8) Balloons Dodge	Avoid the balloons and collect all bubbles.	(19) Letter & Number	Collect a bubble with a letter and one with a number alternately.
(9) Butterflies	Free all the butterflies from the bubbles.	(20) Even Numbers	Collect the bubbles with even numbers.
(10) Colors for Hands	Collect the red and yellow bubbles with the corresponding red and yellow colored hand.	(21) Own Colors*	The front player collects all the yellow bubbles, the back player only collects the green bubbles.
(11) Cross Over	Collect the red and yellow bubbles with the corresponding red and yellow colored hand. This level includes cross over movements.	(22) Alternate Collect*	Always take turns collecting a bubble. One bubble by the back player, one by the front player, etc.

a hot air balloon floats in the air in front of the players and displays the game level task in text form (see Figure 1c). Meanwhile, based on the ROM of each player, soap bubbles or other objects are spawned over the river. Using the ROM points, we create a 2D polygon between these outer limits and the canoe object to define possible spawn locations. We choose random, evenly distributed points inside the polygon to make sure that all generated points are within the player's reach. We can now dynamically modify the difficulty of a level by tuning multiple parameters, depending on the player's success. Different reach difficulty can be achieved by down-scaling the polygon and changing the distance between each of the generated bubbles in the direction the canoe travels. This enables easier, slow-paced, as well as harder levels that require fast decision-making. In the cross over level, we can additionally increase the percentage of bubbles that spawn on the opposite side of the hand that has to collect it.

Depending on the chosen level, different tasks are presented. For example, in the higher number level, players collect the soap bubble with a higher number relative to the bubble partner (see Figure 1c). While traveling along the river, players also pass through caves, which have several vines (see Figure 2c). To finish a level, players perform the following physical movements: torso, reaching, stretching, and leaning (see Figure 1d). At the end of the game, players get a scoreboard where they can see their score and compare it to other players.

4 RESULTS: PROTOTYPING SESSIONS & INTERVIEWS

We used different methods (e.g., observation and field notes, semi-structured interviews) to obtain data in our project. The full list of (translated) questions can be found in the supplementary material. For the game versions and changes used in each session, please see Table 2. All participants are inhabitants from the same local senior living facility. For sessions between I - IV, a healthcare professional was also present.

For the sessions that included interviews, the first author used a thematic analysis method (reflexive orientation) [3–5]. The researcher decided on deductive categories for each session¹. Then

inductively and iteratively coded the interviews under these categories and finally created themes. For the scope of this paper, we only present themes related to *Canoe VR*. We report quotes of the participants based on their specialty in the respective session².

Session I. We aimed to decide on an initial concept of an exergame. Two exergame ideas were implemented as early prototypes: *Canoe VR* and *Flower VR*. Two players tested both games ($M=76$ years, $SD=0$, $female=2$).

Canoe VR was found rich in terms of movement patterns. On a scale from 1 (not at all) to 7 (very much so), the mean enjoyment and exhaustion scores of the players were 6 ($SD=0$) and 1 ($SD=0$). The players and the healthcare professional liked the following aspects: "left and right, recognizing color, head retraction, vines, upper body rotation, movement, points"-HP, "much more movement"-P₁, and "the overall picture of landscapes"-P₂. One player experienced some technical problems due to accidentally pressing a button, which rotated their seating position. At the same time, the health professional reported concerns regarding the risk of falling. They suggested to improve the animal part of the game and add apples of different heights. They rated the enjoyment, presence, and exhaustion of the players as 6, 1, and 1 respectively. The health professional found *Flower VR* unsuitable for fit older adults, because the game flow included too many steps to remember easily. We decided to continue with *Canoe VR*, as it was well received by both the players and the health professional and offers more opportunities to create varying cognitive and physical exercises.

Session II. This session was conducted with two older adults ($M=82.5$ years, $SD=7.78$, $female=2$), who played *Canoe VR*.

The mean enjoyment, presence, and exhaustion scores of players were assessed as 7 ($SD=0$), 6 ($SD=1.41$), and 1 ($SD=0$), respectively. Both players enjoyed the game experience and accompanying music. They liked the following about the game: "the whole game"-P₁ and "that you do something and that it also succeeds and the music was also good. The surroundings also look good, it was so nicely bright"-P₂. They also reported some negative details: "apples left and right sometimes hung too close, hard to tell which apples are in front and back"-P₁. While for P₂, the instructions for feeding animals was too fast and not readable, for P₁, the feeding part was irritating because

¹Session IV: physical activities, games, VR and VR games, and *Canoe VR*. The interviews with physiotherapists: games and VR, physical activities, and *Canoe VR*. Session VI: calibration phase, usability, agents and avatars, multiplayer, and *Canoe VR*.

²PH, PT, HP, P₁ represent a physiotherapist, trainer, and health professional and first player in a session, respectively.

Table 2: This table presents the changes and played game levels in each prototyping session.

SESSIONS	GAME VERSION
SESSION I	In the first version of <i>Canoe VR</i> , the players were placed in a canoe on a river. Their task was to collect all red apples, and avoid the black apples and vines in tunnels (2 minutes). This was explained to them in a short tutorial. There was another level where they could collect fishes jumping out of the river, which had a duration of 45 seconds. At the end, they were asked to feed some animals with the collected food, by throwing the food towards them. The whole game took at least 200 seconds, including the tutorial. Background music accompanied the players. In <i>Flower VR</i> , players were placed in a garden scene. The task was to recreate a certain bouquet with different kinds of flowers that was shown to them. They had to pick the correct seeds, plant them in a flowerpot, water them, and afterwards place them in a vase.
SESSION II	We increased the length of the game to 220 meters (240 including the tutorial) and excluded the fishing part. At the end, the food was automatically given to the animals and did not have to be thrown anymore.
SESSION III	We introduced yellow and green apples in addition to red ones. The ending concept was changed to having a picnic with the collected apples. In three different levels, the players' task was to collect i) only red apples (60 seconds), ii) all red apples and 2-5 (fixed random number) yellow and 2-5 (fixed random number) green apples (90 seconds), and iii) the apple with a higher number (60 seconds).
SESSION IV	We made the second level easier by removing the green apples. The player played the levels i) only red apples, ii) all red and 2-5 (fixed random number) yellow apples, iii) the apple with the higher number, iv) alternately colored apples, and v) a level with soap bubbles. Each level was played for 60 seconds and the total game time was at least 280 seconds, including a tutorial.
SESSION V	This session featured <i>Canoe VR</i> on the Meta Quest 2 for the first time, with a new scene (see Figure 1a) and hand tracking instead of controllers. ROM calibration was added where the players sat straight and had to stretch their arms to the front as far as possible. The apple trees were exchanged for freely floating soap bubbles of different colors to enable more flexible positioning of the objects. Three levels were presented, with a length of 120 seconds each: all good, butterflies, and collecting red and yellow bubbles alternately. At the end of the level, a scoreboard was presented that showed how many bubbles were collected and how many vines were touched.
SESSION VI	Two game versions were presented: A single player version with a virtual agent as a second player, and a multiplayer version where two users could play together. The second player/agent was represented with two different avatars (changed during the game play) which had either i) only trunk and hands (see Figure 2a), or ii) full body (see Figure 2c). In the agent version, an agent was sitting in the canoe in front of the player, performing the tasks with them (see Figure 2a). The calibration was changed to a more playful solution, where the users had to push four balloons (two for each hand, one above them and one on the side) as far away as possible (see Figure 1b). Each balloon had to be touched at least once to finish the calibration. The following four levels were played for 60 seconds with a spawn distance of 1 meter between objects: fruits, butterflies, dodge balloons, and colors for hands. In the multiplayer version, the two players were sitting behind each other in the canoe (see Figure 2b,c). The played levels were fruit, own colors, alternate collect, all bubbles.
SESSION VII	The agent was removed in this version. The following four levels were played, with a duration of 60 seconds each, and a start distance of 1.5 m between collectible items: butterflies, colors for hands, cross over, and fruit. The distance between bubbles was evaluated every 10 bubbles and adjusted automatically by 0.2 m if more than 80 % of bubbles were correct or more than 40 % were wrong.
SESSION VIII	A remote was used to control the game flow. At the start, a tutorial asked the players to collect three bubbles while the canoe was still stationary to help them understand the task. We switched back from hand tracking to using controllers because of a more stable tracking. The game featured six levels, which were played in a random order and with different start distances between bubbles, 60 seconds each: all good (distance = 2 m), music (distance = 2 m), even numbers (distance = 1.2 m), letter and number (distance = 2 m), cross over (distance = 1.2 m), higher number (distance = 1.2 m). Distance between bubbles was evaluated every 5 meters and adjusted by 0.3 m if more than 80 % were correct or more than 20 % were wrong. The high score presented the top 10 scores and always included the players' score.

the basket was transferred to the land and the player stayed in the canoe.

Session III. This session was conducted with older adults with dementia ($N=3$, $M=81.33$ years, $SD=10.79$, $female=3$).

Two players answered the enjoyment question with the following words: “*flawless*” and “*nice*”. For presence, the answers were: “*yes*” and “*yes sure*”, and “*no*”. For exhaustion, a player answered with “*a little*”, and the other with “*no*”. Two players liked the task of collecting apples. The other player indicated that they liked the diversity with apples. Overall, the game level with all red, and a fixed amount of yellow and green apples was found too difficult. The health expert reported: for the enjoyment (“*well liked 5-6 [as a score]*”) and exhaustion (“*no*”). They indicated that players had difficulties to understand and hear the instructions. They suggested to make the tutorial slower, avoid giving too much information, and to highlight the vines under the bridges more clearly or to use spiderwebs instead.

Session IV. Six older adults ($M=78.83$ years, $SD=3.06$, $female=5$, $male=1$) were recruited. We aimed to gain a deeper understanding about our user group, and, therefore, conducted semi-structured interviews with them.

Theme: Canoe VR is very well received, but the preferences of players for game objects and levels varied. Most players enjoyed the game: “*Yes, I like it very much*”- P_5 . The cognitive and physical value of the game was appreciated by some players: “*That you just always keep moving and always have to pay attention, then duck, again pay attention to the apples etc. [...] I thought it was very nice, very interesting*”- P_1 . But also: “*That really challenged me [smirks]*”- P_4 . All players found the length of the overall game appropriate, however, P_2 player commented on having shorter individual levels and quick changes between them. Some players mentioned that they liked the apples more compared to soap bubbles, while only one reported a reason: “*Yeah, was more representational somehow*”- P_3 . We speculate that the preference for apples could be due to the

variety in apple tasks. There was subjectivity for the best task: all red and exactly [2-5] yellow apples, the higher number, combination of all these different things, and collecting red and yellow apples alternately. All players were positive to play the game regularly with some task variations: “Yes, I can imagine that”-P₃.

Interview with Physiotherapists. After the positive feedback in the last session, we decided to present the game to physiotherapists. This semi-structured interview was conducted in a video call attended by two experts ($M=46$ years, $SD=7.07$, $female=1$, $male=1$). During the interview, we presented a video from the last prototyping session with an older adult playing the game and showed them some concept pictures.

Theme: Canoe VR features rich movements and can offer valuable options (being mobile and social value), but it should be realistic and familiar. Both therapists found the featured movements very good: “That you can also lower the arm and work with the trunk, raise the arm again, that I also have pause phases in between and not constantly a tension, a contraction of a muscle group. Because that is much too strenuous in the long run. [...] a game like that is really super”-PH₂. The idea of having the game portable is found interesting because users can easily try out the game. A multiplayer mode to facilitate social interaction was also well received (“when we interact as a team. [...] There comes then also again an added value to it”-PH₂). Both experts strongly emphasized the need for graphical realism, in particular, consistency with the real world: “Not that the graphics have to be designed in a great way, but I wouldn’t keep it abstract. [...] But I wouldn’t overburden [...] the participants with disturbing things [...] where everything is unrealistic. Or where the apple is not red and yellow, but blue and purple... So I would go very close to reality, especially with the graphics”-PH₂.

Session V. This session was conducted in the gym of the facility with an attendance of a physiotherapist and a trainer ($N=2$, $M=41.5$ years, $SD=0.71$, $female=1$, $male=1$). We asked the older adults ($N=7$, $M=83.71$ years, $SD=3.25$, $female=6$, $male=1$) in the gym if they would like to play *Canoe VR*. The experts also played the game to give more precise feedback.

Canoe VR was found enjoyable by the players ($M=3.81$, $SD=0.50$, KIM [38] enjoyment/interest sub-scale, *not at all true/0 – very true/4*). The players felt low levels of exhaustion ($M=1.57$, $SD=0.98$, a single custom item, *not exhausted at all/1 – very exhausted/5*). Most players also gave positive comments: “overall, an interesting, movement-intensive game”-P₆. However, P₄ reported technical issues such as the font becoming blurry and quickly disappearing. Also, for P₇, the game was not something they would play every day. For the experts, the use of VR for older adults’ exercises is considered positive: “very good approach to apply digital movement offers in training therapy, diversified, innovative, motivating for all difficulty levels”-PH. Both also rated the enjoyment of the players positively ($M=4.5$, $SD=0.71$, a single custom item, *very bad/1 – very good/5*). The trainer reported that the VR headset “[...] size partly too big”, however, for the therapist the usability of the system was “very good, [...] could be individualized if necessary, otherwise good flow, smooth gameplay, pleasant design ambiance”. The movements ($M=4$, $SD=0$) and their daily functionality for older adults ($M=4$, $SD=0$) were rated high (a single custom item, *very bad/1 – very good/5*). While the trainer found the bubbles a bit too fast for the

players, the physiotherapist suggested to have “shorter sequences, lower movements [...], dynamic, sitting “actively” at the front of the chair, cross-over movements (of the hands oriented to the center of the body)”.

Session VI. In this session, we focused on the social value of the game. We tested a version with older adults ($M=78.75$ years, $SD=4.71$, $female=6$, $male=2$) playing with a virtual agent ($N=4$), and a multiplayer version with two players playing together ($N=4$). We took observation, field notes, and interviewed the players in groups of one, three, and four players.

Theme: The sitting configuration and the agent were not received well, but overall players reported positive player experience with consideration of some aspects. All players playing with the agent noticed the agent, however, it faded out after some time: “I only saw them for a short moment and then I didn’t see them anymore”-P₃. P₁ explicitly described the changes in the appearance of the avatars, however, most of these players did not see a need for an agent: “actually, they can go away”-P₂. Similarly, P₄ also realized the changes, but they focused more on the task: “At the beginning I saw that, but then I was so concentrated on [task] what concerns me [...]”-P₄. In the multiplayer mode, players mostly indicated that they enjoyed to play in pairs, while a player, who played with the agent negatively, commented about playing with a real person “[...] you would concentrate on that person and [...] would be too distracted”-P₂. However, a front player felt like they played alone: “I did not notice my [partner]”-P₇.

Overall, many players indicated positive player experience: “Somehow you had the feeling that you were sitting in the boat. Then the waves moved. So I thought that was actually quite nicely done”-P₆ and “You couldn’t just sit there and watch, but it was a challenge to [perform the tasks]. So I found that quite good”-P₄. Many players preferred the avatar with the full body representation while for P₄, it did not matter as the task was more important. Regarding the design, a few players had difficulties to see the butterflies, required some time to understand the thumbs up gesture, or had issues with the behavior of the calibration balloons: “So I had the impression, the balloons do not really react when you push them away”-P₈. Some players suggested to increase the length of the gameplay.

Session VII. This session was held in the same gym as Session V, and with the same fitness experts. Three older adults ($M=84.33$ years, $SD=1.53$, $female=2$, $male=1$) and the experts played *Canoe VR* (see Figure 2d).

The players rated their enjoyment of the game as high ($M=3.89$, $SD=0.19$, KIM). However, their exhaustion was low ($M=1$, $SD=0$). For P₃, “overall, the game is interesting, challenging, concentration and movement enhancing”. While P₁ suggested to have the texts visible for a longer time, P₂ suggested to improve the calibration sequence: “Slightly change the setting at the start so that the balloons do not fly so far (away)”. According to the experts, the older players enjoyed the game ($M=4.5$, $SD=0.71$). For usability, the physiotherapist commented on a safety issue: “possibly too wide cross movements (inclinations to fall)”. The evaluation of the movements patterns ($M=4.5$, $SD=0.71$) and their functionality ($M=4.5$, $SD=0.71$) for daily life of older adults were high. The butterfly level caused a little challenge: “visibility of the “butterflies” sometimes a bit difficult”-PH, as also mentioned by P₁. Both experts found the game valuable:

“variety is good, timing is ideal, cross movements is extremely valuable therapeutically, good trunk activity (rotation/trunk extension)”-PH. For further improvements, they suggested: *“more stretching movements of the torso (arms stretching upwards)”-PH*, *“reduce the range of motion of insecure people if necessary [...] fun of movements in the game is given, if necessary sound as confirmation of the achieved movement”-PH*, and *“Bonus round! “Special effect” if you made the quest (e.g., fireworks or a lovely melody), optional music”-PT.*

Session VIII. We conducted this session with people with dementia ($N=5$, $M=81.2$ years, $SD=8.58$, $female=4$, $male=1$). We focused on three points: interaction with the game after the improvements, understanding how challenging the game is, and finding a suitable starting difficulty. To determine the difficulty, we adjusted the distance between bubbles using a modified version of the staircase method [9]. The levels started with either 1.2 meters or 2 meters distance between bubbles and adjusted itself every 5 meters by 0.3 meters, depending on game performance within this section.

The players played an average of 3.8 ($SD=1.79$, $Min-Max=1-6$) levels, based on their own wishes. During the game, some difficulties were observed. For example, P_1 had challenges to complete the tutorial. Since we did not implement a way to skip the tutorial on the remote control, it was turned off for the other four players. Also, none of the participants was able to finish the calibration sequence. When they had tried their best and did not make any more progress, we skipped the calibration on the remote control, playing with the default values. With varying physical and cognitive abilities, their success in the game also changed from person to person, e.g., for P_5 , the game worked quite smoothly while P_1 had some difficulties. With the dynamic difficulty, we found that the mean distance between bubbles (mean of all data points) approached 2.38 meters ($SD=0.91$). We note that players played different levels with varying cognitive and physical tasks and a different amount of levels, therefore, the average distance serves as a reference start point for the future, not for comparison purposes.

5 DISCUSSION & CONCLUSION

In this work, we presented and evaluated *Canoe VR*, which enables cognitive and physical training for older adults. We considered cognitive variances as well as physical needs of older adults. In an HCD process [12], we gathered feedback from older adults with and without dementia, a health professional, and fitness experts. We iteratively improved the game to provide positive player experience and offer both cognitive and physical training opportunities.

We used various methods in this study for data collection. Although, this can be seen as a limitation in terms of not obtaining homogeneous findings, it also offers rich results. Our sample size is low and the results should be interpreted with this in mind. Also, the reflexive orientation of thematic analysis requires the deep engagement of the researcher with data, therefore does not require more than one coder [3–5]. Following qualitative research practices [26], we report on reflexivity that might have led to bias in the analysis, such as the main author’s prior research experience with older adults, VR game experience, and a background in psychology and cognitive systems. Yet, these also stands as a strong ground to understand and interpret the data.

Overall, the results indicate that the game was very well received by users and the movements were found rich by the experts. Although participants suggested improvements or pointed to unsuccessful points as well, they generally expressed positive experiences. Our results, aligning with prior work [6], show that older adults’ VR exergame experiences were positive, in particular during COVID-19. For this specific game scenario, the used virtual agent was not found to be helpful. In contrast, a multiplayer mode was a good addition, but the front players also desired more interactions with their partner, leaving a gap for improvement.

Canoe VR was developed through a design process with multiple stakeholders and several prototyping sessions. We also showed that this game can be used in regular gym sessions of older adults. By building on previous work [18], we show that a VR exergame can be suitable as an additional fitness tool for this user group, yet further scientific explorations are required to generalize this result. Although the game features a variety of cognitive tasks and movements, more research is needed to examine the cognitive and physical benefits (e.g., using subjective and objective measures) of this designed game.

In terms of varying abilities in the population, our findings indicate that *Canoe VR* offers opportunities for both, players with and without dementia. The game settings provide options to change the game tasks, elements, length, and difficulty. Further, dynamic difficulty integration allowed the players with varying abilities to experience the game and exercise. Therefore, based on the sessions with players with dementia, we argue that this game also matches their abilities. The importance of this is also emphasized in [16]. However, we particularly plan to explore this game in more detail with both groups in our future work. We plan to test this game with older players with and without dementia in a long-term study to understand which aspects of the game are suitable for which group and how these affect the game and their cognitive abilities.

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